

Education

- 2025–now **Postdoctoral Researcher (Advisor: Ernst Kuwert)**, *University of Freiburg*, Germany
- 2023–2025 **Postdoctoral Researcher (Advisor: Gang Tian)**, *Peking University (BICMR)*, China
- 2015–2023 **Ph.D. in Mathematics (Advisor: Jiayu Li)**, *University of Chinese Academy of Sciences (AMSS, CAS)*, China
- 2011–2015 **B.Sc. in Mathematics**, *Wuhan University*, China

Research Interests

Geometric analysis

Publications

Published

- [1] Yuchen Bi, Jiayu Li, “The Prescribed Q -Curvature Flow for Arbitrary Even Dimension in a Critical Case”, *Advances in Mathematics* 438 (2024), 109478.
- [2] Yuchen Bi, Jiayu Li, Lei Liu, Shuangjie Peng, “The C^0 -convergence at the Neumann boundary for Liouville equations”, *Calculus of Variations and Partial Differential Equations* 62 (2023), no. 3, Paper No. 107.
- [3] Yuchen Bi, Jie Zhou, “Optimal rigidity estimates for varifolds almost minimizing the Willmore energy”, *Transactions of the American Mathematical Society* 378 (2025), no. 2, 943–965.

Preprints

- [4] Yuchen Bi, Jintian Zhu, Positive Scalar Curvature Obstructions via Singular Dimension Descent”, arXiv:2606.20528.
- [5] Yuchen Bi, Jintian Zhu, Curvature-free Effects from Volume Growth and Ends-counting and Their Applications”, arXiv:2605.12403.
- [6] Yuchen Bi, Jintian Zhu, Riemannian Penrose Inequality in All Dimensions”, arXiv:2605.00680.
- [7] Yuchen Bi, Tianze Hao, Shihang He, Yuguang Shi, Jintian Zhu, A Proof for the Riemannian Positive Mass Theorem up to Dimension 19”, arXiv:2603.02769.
- [8] Yuchen Bi, Jie Zhou, Linear Quantitative Rigidity for Almost-CMC Surfaces”, arXiv:2601.09457.

- [9] Yuchen Bi, Jie Zhou, Quantitative Stability of the Clifford Torus as a Willmore Minimizer", arXiv:2511.19681.
- [10] Yuchen Bi, Jintian Zhu, Mass-Capacity Inequality Modeled on Conformally Flat Manifolds", arXiv:2511.13215.
- [11] Yuchen Bi, Jie Zhou, Bi-Lipschitz Regularity of 2-Varifolds with the Critical Allard Condition", arXiv:2212.03043.
- [12] Yuchen Bi, Jie Zhou, "Bi-Lipschitz Rigidity for L^2 -Almost CMC Surfaces", arXiv:2212.02946.

Selected Talks

- 05/2026 "Linear Quantitative Rigidity for Almost-CMC Surfaces", 43rd S"uddeutsches Kolloquium "uber Differentialgeometrie, University of T"ubingen. Joint work with Jie Zhou.
- 03/2026 "Quantitative Rigidity of the Clifford Torus as the Willmore Minimizer", 2026 Xiamen International Conference on Geometric Analysis. Joint work with Jie Zhou.
- 03/2026 "Mass-Capacity Inequality Modeled on Conformally Flat Manifolds", 2026 Geometric Analysis Spring Conference, University of Science and Technology of China, Hefei. Joint work with Jintian Zhu.

Teaching Experience

- 2026 **Teaching Assistant**, *University of Freiburg*, Germany
Curves and Surfaces, Summer Semester 2026, with Ernst Kuwert.
- 2025–2026 **Lecture and Exercise Sessions**, *University of Freiburg*, Germany
Graduate course "Differential Geometry", Winter Semester 2025/26.